

Grade 10 Chemistry Unit 1 Chemical Reaction and Stoichiometry

At the end of this unit, you will be able to

- define the basics of chemical reaction;
- describe the four major types of reactions;
- develop skills in writing and balancing chemical equations;
- define oxidation - reduction reactions analyse redox reactions by specifying the oxidizing agent, the reducing agent, the substance reduced or oxidized;
- explain molecular mass, formula mass, molar mass, empirical and molecular formulas;
- determine molecular mass, formula mass, molar mass, empirical and molecular formulas
- describe the mole concept
- solve problems related to moles of substances
- develop skills in solving problems based on chemical equations (mole-mole, mass - mass, volume - volume and mass - volume problems);
- develop skills in determining the limiting reactant, theoretical yield, actual yield and percentage yield;

1.1 Introduction

At the end of this section, students will be able to

- define chemical reaction;
- give some examples of chemical reactions.
- explain physical and chemical changes using examples

1.2 Chemical Equations

At the end of this section, students will be able to

- explain the conventions used to write chemical equation
- write the chemical equations of varieties of chemical reactions;
- balance chemical equations using;
 - The Inspection method,
 - The Least Common Multiple (LCM) method,
 - The Algebraic method.

1.3 Types of Chemical Reactions

- the end of this section, students will be able to
- list the four types of chemical reactions;
- define combination reaction and give examples;
- conduct experiments on combination reactions in groups;
- define decomposition reaction and give examples;
- conduct some experiments on decomposition reactions in groups;
- define single displacement reaction and give examples;
- conduct some experiments on simple displacement reactions in groups;
- define double decomposition reaction and give examples;
- conduct some experiments on double displacement reactions in group.

1.4 Oxidation and Reduction

at the end of this section, students will be able to

- define redox reactions;
- define the terms oxidation and reduction in terms of electron transfer;
- define oxidation number (oxidation state);
- state oxidation number rules.;
- determine the oxidation number of an element in a given formula.

1.4.1 Oxidation Number or Oxidation State

At the end of this section, students will be able to

- define oxidation number (oxidation state);
- state oxidation number rules.;
- determine the oxidation number of an element in a given formula.

1.4.2 Oxidizing and Reducing Agents

At the end of this section, students will be able to

- define the terms reducing and oxidizing agent;
- identify reducing and oxidizing agents from a given redox reaction;
- compare and contrast oxidizing and reducing agents.

1.4.3 Analyzing Redox Reactions

- At the end of this section, students will be able to explain why every reduction reaction is accompanied by oxidation reaction.

1.4.4 Balancing Redox Reactions: Oxidation-Number-Change Method

- At the end of this section, students will be able to balance redox reactions using oxidation change method.

1.5 Molecular and Formula Masses, the Mole Concept and Chemical Formulas

At the end of this section, students will be able to

- explain and determine molecular mass, formula mass, molar mass, empirical and molecular formulas;
- describe the mole concept and solve problems related to moles of substances.

1.5.1 Molecular mass (MM) and Formula Mass (FM)

- At the end of this topic, students will be able to explain and determine molecular mass, formula mass.

1.5.2 The Mole Concept

At the end of this topic, students will be able to

- define mole and explain the concept using examples;
- define molar mass;
- calculate the molar masses of substances.

1.5.3 Chemical Formulas - Empirical and Molecular Formulas

At the end of this topic, students will be able to

- define empirical and molecular formulas;
- determine empirical and molecular formulas;
- calculate percentage composition of an element in a substance.

1.6 Stoichiometry

At the end of this section, students will be able to

- deduce mole ratios from balanced chemical equations;
- solve mass-mass problems based on the given chemical equation;
- define molar volume;
- state Avogadro's principle;
- solve mass-mass problems based on the given chemical equation;
- solve mass -volume problems based on the given chemical equation;
- solve volume -volume problems based on the given chemical equation;
- describe limiting and excess reactants.